Assignment 5: ViT-S Fine-tuning with LoRA and Adversarial Attacks

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1 Introduction

This report presents the results of Assignment 5, which consists of two main tasks:

- **Q1:** Fine-tuning a pre-trained ViT-S (Vision Transformer Small) model on CIFAR-100 using LoRA (Low-Rank Adaptation) with various hyperparameter configurations.
- **Q2:** Implementing adversarial attacks (FGSM, PGD, BIM) on CIFAR-10 using IBM Adversarial Robustness Toolbox (ART) and training adversarial detectors.

All experiments were conducted using Docker containers on GPU-1 (48GB VRAM).

1.1 Links

- **WandB Q1:** `https://wandb.ai/m25csa010-iit-jodhpur/assignment5-q1`
- **WandB Q2:** `https://wandb.ai/m25csa010-iit-jodhpur/assignment5-q2`
- **HuggingFace Model:** `https://huggingface.co/JD16112001/vit-lora-cifar100-best`
- **GitHub Repository:** `https://github.com/Jyoti-Dwivedi-010/MLOps-Jyoti-Dwivedi-M25C tree/Assignment-5`

2 Q1: ViT-S Fine-tuning with LoRA on CIFAR-100

2.1 Methodology

We fine-tuned a ViT-S model pre-trained on ImageNet for CIFAR-100 classification (100 classes). The experiments include:

1. **Baseline:** Fine-tuning only the classification head (no LoRA)
2. **LoRA Grid Search:** Applying LoRA to Q, K, V attention weights with:

- Ranks: 2, 4, 8
- Alpha: 2, 4, 8
- Dropout: 0.1

3. **Optuna Search:** Hyperparameter optimization for LoRA configuration
4. [**Optional**] **Partial Freeze:** LoRA on frozen layers with last 2 blocks trainable

2.2 Baseline Results (Without LoRA)

Table 1: Baseline Training (Classification Head Only, No LoRA)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	1.6890	0.9144	0.6389	0.7690
2	0.7415	0.7601	0.7995	0.7880
3	0.6319	0.7104	0.8216	0.7962
4	0.5771	0.6884	0.8341	0.8004
5	0.5395	0.6691	0.8423	0.8038
6	0.5073	0.6560	0.8503	0.8064
7	0.4871	0.6571	0.8554	0.8060
8	0.4703	0.6514	0.8570	0.8058
9	0.4523	0.6442	0.8646	0.8068
10	0.4395	0.6439	0.8666	**0.8116**

Baseline Test Results:

- Best Validation Accuracy: **81.16%**
- Trainable Parameters: **38,500**
- Total Parameters: 21,704,164

2.3 LoRA Experiment 1: Rank=2, Alpha=2

Table 2: Experiment 1: Rank=2, Alpha=2, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.5685	1.0576	0.4805	0.7882	0.5181
2	0.6774	0.6015	0.8365	0.8464	0.8790
3	0.4721	0.4972	0.8680	0.8612	0.9675
4	0.4042	0.4510	0.8810	0.8672	1.0030
5	0.3665	0.4198	0.8893	0.8764	1.0285
6	0.3363	0.4024	0.8974	0.8810	1.0252
7	0.3173	0.3885	0.9020	0.8836	1.0281
8	0.2976	0.3774	0.9101	0.8850	1.0251
9	0.2831	0.3736	0.9108	0.8864	1.0382
10	0.2702	0.3621	0.9145	**0.8898**	1.0333

2.4 LoRA Experiment 2: Rank=2, Alpha=4

Table 3: Experiment 2: Rank=2, Alpha=4, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.4743	0.9674	0.4966	0.7984	0.8973
2	0.6352	0.5735	0.8425	0.8508	1.4228
3	0.4550	0.4769	0.8711	0.8646	1.5363
4	0.3916	0.4355	0.8841	0.8706	1.5776
5	0.3554	0.4076	0.8918	0.8794	1.5983
6	0.3256	0.3906	0.8994	0.8820	1.5840
7	0.3071	0.3794	0.9040	0.8874	1.5914
8	0.2879	0.3682	0.9123	0.8872	1.5741
9	0.2737	0.3657	0.9145	0.8890	1.5928
10	0.2610	0.3562	0.9170	**0.8946**	1.5902

2.5 LoRA Experiment 3: Rank=2, Alpha=8

Table 4: Experiment 3: Rank=2, Alpha=8, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.3712	0.8822	0.5153	0.8090	1.5540
2	0.6015	0.5489	0.8471	0.8534	2.2968
3	0.4405	0.4604	0.8747	0.8668	2.4102
4	0.3805	0.4248	0.8867	0.8724	2.4361
5	0.3448	0.3989	0.8947	0.8794	2.4517
6	0.3155	0.3837	0.9018	0.8818	2.4153
7	0.2977	0.3735	0.9073	0.8890	2.4271
8	0.2786	0.3629	0.9142	0.8876	2.3890
9	0.2638	0.3626	0.9177	0.8890	2.3900
10	0.2514	0.3550	0.9198	**0.8934**	2.4132

2.6 LoRA Experiment 4: Rank=4, Alpha=2

Table 5: Experiment 4: Rank=4, Alpha=2, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.5801	1.0627	0.4791	0.7884	0.3644
2	0.6779	0.5907	0.8378	0.8466	0.6268
3	0.4725	0.4916	0.8680	0.8634	0.6973
4	0.4020	0.4436	0.8822	0.8688	0.7234
5	0.3647	0.4172	0.8901	0.8780	0.7413
6	0.3328	0.3965	0.8984	0.8814	0.7421
7	0.3119	0.3819	0.9046	0.8824	0.7402
8	0.2960	0.3767	0.9069	0.8850	0.7442
9	0.2775	0.3647	0.9126	0.8882	0.7337
10	0.2622	0.3615	0.9182	**0.8876**	0.7273

2.7 LoRA Experiment 5: Rank=4, Alpha=4

Table 6: Experiment 5: Rank=4, Alpha=4, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.4863	0.9698	0.4955	0.7988	0.6351
2	0.6339	0.5600	0.8430	0.8522	1.0261
3	0.4536	0.4708	0.8714	0.8660	1.1071
4	0.3881	0.4270	0.8856	0.8752	1.1362
5	0.3522	0.4045	0.8935	0.8786	1.1518
6	0.3209	0.3847	0.9011	0.8828	1.1439
7	0.3006	0.3715	0.9076	0.8854	1.1423
8	0.2846	0.3684	0.9102	0.8870	1.1470
9	0.2667	0.3559	0.9154	0.8892	1.1267
10	0.2511	0.3548	0.9218	**0.8892**	1.1113

2.8 LoRA Experiment 6: Rank=4, Alpha=8

Table 7: Experiment 6: Rank=4, Alpha=8, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.3824	0.8876	0.5140	0.8058	1.1061
2	0.5982	0.5330	0.8484	0.8572	1.6569
3	0.4365	0.4507	0.8749	0.8688	1.7251
4	0.3747	0.4134	0.8883	0.8786	1.7482
5	0.3395	0.3932	0.8972	0.8802	1.7590
6	0.3089	0.3756	0.9040	0.8856	1.7359
7	0.2889	0.3624	0.9110	0.8862	1.7280
8	0.2724	0.3614	0.9128	0.8890	1.7250
9	0.2548	0.3489	0.9197	0.8910	1.6879
10	0.2388	0.3491	0.9257	**0.8918**	1.6586

2.9 LoRA Experiment 7: Rank=8, Alpha=2

Table 8: Experiment 7: Rank=8, Alpha=2, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.5661	1.0633	0.4825	0.7896	0.2529
2	0.6745	0.5967	0.8363	0.8456	0.4408
3	0.4709	0.4949	0.8671	0.8604	0.4902
4	0.4022	0.4507	0.8812	0.8684	0.5109
5	0.3622	0.4180	0.8905	0.8756	0.5087
6	0.3326	0.3970	0.8997	0.8812	0.5121
7	0.3129	0.3828	0.9046	0.8830	0.5194
8	0.2947	0.3722	0.9078	0.8876	0.5189
9	0.2796	0.3664	0.9128	0.8894	0.5266
10	0.2647	0.3585	0.9161	**0.8908**	0.5231

2.10 LoRA Experiment 8: Rank=8, Alpha=4

Table 9: Experiment 8: Rank=8, Alpha=4, Dropout=0.1

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.4732	0.9692	0.5001	0.7990	0.4368
2	0.6300	0.5676	0.8433	0.8514	0.7135
3	0.4516	0.4761	0.8705	0.8646	0.7721
4	0.3886	0.4361	0.8841	0.8724	0.7932
5	0.3495	0.4047	0.8942	0.8788	0.7843
6	0.3206	0.3863	0.9024	0.8852	0.7803
7	0.3010	0.3741	0.9072	0.8872	0.7929
8	0.2833	0.3631	0.9112	0.8908	0.7900
9	0.2678	0.3585	0.9157	0.8918	0.8053
10	0.2526	0.3512	0.9206	**0.8934**	0.7946

2.11 LoRA Experiment 9: Rank=8, Alpha=8 (Best Configuration)

Table 10: Experiment 9: Rank=8, Alpha=8, Dropout=0.1 **(Best)**

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	2.3692	0.8845	0.5164	0.8058	0.7546
2	0.5927	0.5403	0.8486	0.8562	1.1349
3	0.4338	0.4610	0.8740	0.8664	1.1933
4	0.3748	0.4257	0.8873	0.8760	1.2026
5	0.3363	0.3958	0.8978	0.8816	1.1830
6	0.3073	0.3786	0.9056	0.8858	1.1766
7	0.2872	0.3692	0.9103	0.8888	1.1863
8	0.2693	0.3573	0.9150	0.8896	1.1789
9	0.2532	0.3546	0.9195	0.8904	1.1930
10	0.2376	0.3460	0.9255	**0.8932**	1.1801

2.12 Q1 Test Results Summary

Table 11: Q1 Complete Test Results Table

LoRA	Rank	Alpha	Dropout	Test Acc	Trainable Params
without	-	-	-	81.16%	38,500
with	2	2	0.1	88.97%	113,864
with	2	4	0.1	89.20%	113,864
with	2	8	0.1	89.27%	113,864
with	4	2	0.1	88.99%	150,728
with	4	4	0.1	89.06%	150,728
with	4	8	0.1	89.26%	150,728
with	8	2	0.1	89.11%	224,456
with	8	4	0.1	89.27%	224,456
with	**8**	**8**	**0.1**	**89.33%**	**224,456**

Key Observations:

- LoRA significantly improves accuracy from 81.16% (baseline) to up to 89.33%
- Higher rank and alpha generally lead to better performance
- Best configuration: Rank=8, Alpha=8 achieves **89.33%** test accuracy
- Trainable parameters increase from 38,500 (baseline) to 224,456 (r8, a8), still only 1% of total parameters

2.13 Optuna Hyperparameter Search Results

Table 12: Optuna Best Hyperparameters

Parameter	Best Value
Rank	4
Alpha	4
Learning Rate	0.00048
Validation Accuracy	**89.46%**

2.14 [Optional] Partial Freeze Experiment

Configuration: Keep last 2 transformer blocks fully trainable (no LoRA), apply LoRA only to frozen blocks (0-9).

Table 13: Partial Freeze Training (Rank=4, Alpha=4, 2 Trainable Blocks)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc	Grad Norm
1	0.6256	0.4269	0.8266	0.8746	0.7969
2	0.2766	0.4181	0.9121	0.8762	0.7310
3	0.1804	0.4562	0.9409	0.8746	0.6363
4	0.1304	0.4699	0.9573	0.8784	0.6061
5	0.0985	0.5010	0.9677	0.8760	0.5813
6	0.0851	0.5185	0.9714	0.8788	0.5847
7	0.0724	0.5339	0.9754	**0.8818**	0.5695
8	0.0702	0.5580	0.9770	0.8790	0.5793
9	0.0598	0.5632	0.9813	0.8794	0.5389
10	0.0497	0.6144	0.9836	0.8754	0.5287

Table 14: Partial Freeze vs Full Freeze Comparison

Experiment	Trainable Blocks	LoRA Blocks	Test Acc	Trainable Params
Full Freeze + LoRA	0	12	89.06%	150,728
Partial Freeze + LoRA	2	10	88.40%	3,648,868

Analysis: The partial freeze approach achieved 88.40% test accuracy compared to 89.06% for full freeze with LoRA, despite having 24x more trainable parameters (3.6M vs 150K). This suggests that:

- LoRA on frozen layers is more parameter-efficient
- Pre-trained representations in the last layers are already well-suited for the downstream task
- Full-freeze with LoRA maintains better generalization

2.15 Q1 Visualizations

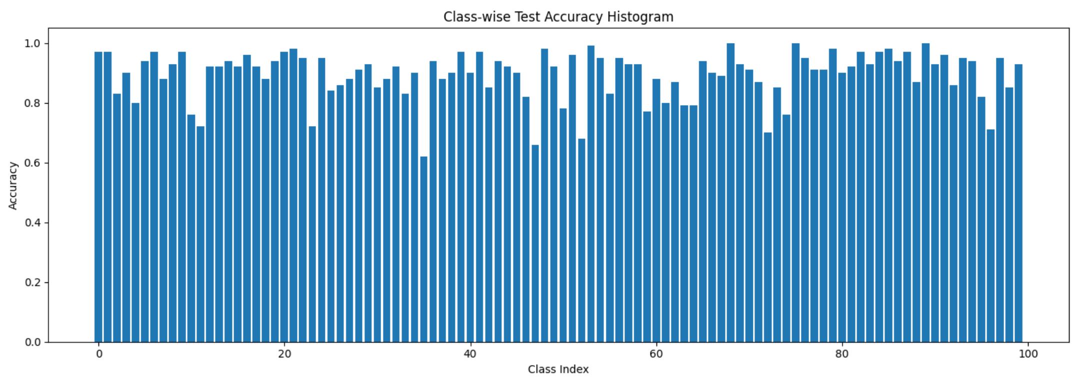

Figure 1: Class-wise Test Accuracy Histogram for Best Configuration (Rank=8, Alpha=8)

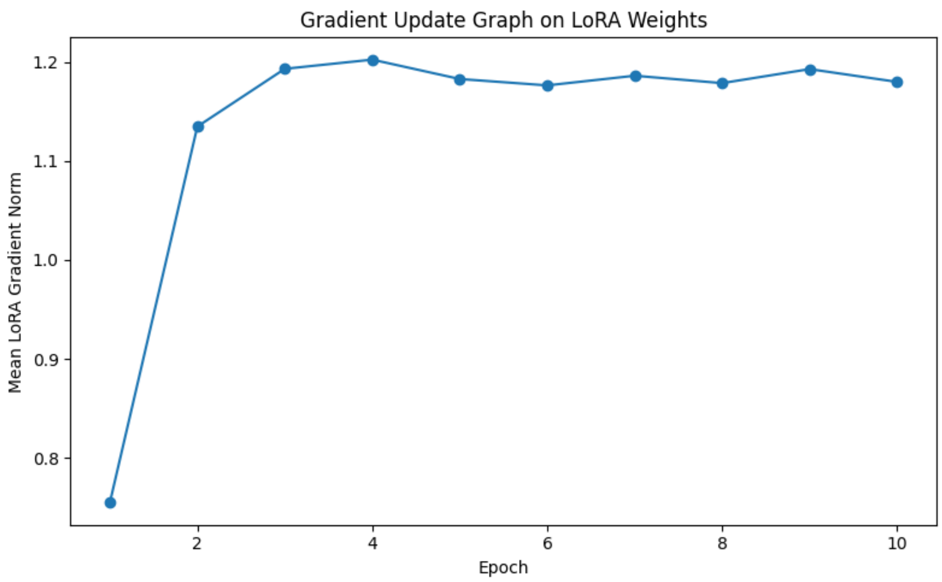

Figure 2: Gradient Update Graph on LoRA Weights During Training

3 Q2: Adversarial Attacks using IBM ART

3.1 Q2(i): FGSM Attack Comparison

3.1.1 Clean ResNet18 Training

We trained a ResNet18 model from scratch on CIFAR-10, achieving the required ≥72% accuracy.

Table 15: Clean ResNet18 Training on CIFAR-10 (Selected Epochs)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	2.3237	2.7658	0.2703	0.3140
5	1.1844	1.3193	0.5745	0.5262
10	0.8449	1.4685	0.7047	0.5270
15	0.6634	0.8048	0.7663	0.7160
20	0.5202	0.7729	0.8188	0.7438
25	0.3808	0.5369	0.8649	0.8150
30	0.2992	0.4796	0.8963	**0.8376**

Final Test Accuracy: 83.26% (Target: ≥72%)

3.1.2 FGSM Attack Results

Table 16: FGSM Attack Comparison (Epsilon = 0.03)

Metric	Clean	FGSM Scratch	FGSM ART
Accuracy	83.26%	9.59%	19.06%
Accuracy Drop	-	73.67%	64.20%

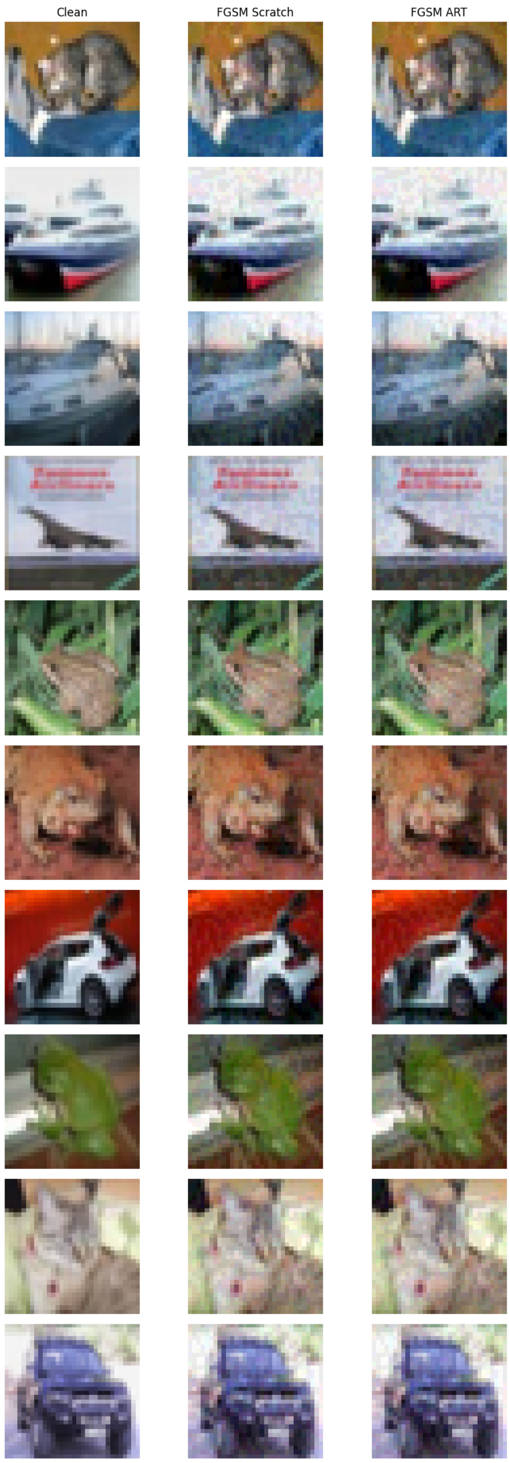

Figure 3: Visual Comparison of Clean and Adversarial Images (FGSM)

Analysis:

- Both FGSM implementations successfully fool the model
- FGSM from scratch achieves a stronger attack (9.59% accuracy vs 19.06% for ART)
- The difference may be due to implementation details (gradient normalization, clipping strategies)

- Both attacks demonstrate the vulnerability of neural networks to adversarial perturbations

3.2 Q2(ii): Adversarial Detection Models

3.2.1 PGD Detector Training

Table 17: PGD Adversarial Detector Training (ResNet-34)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7145	0.6976	0.5013	0.5049
3	0.5511	0.6947	0.6793	0.5555
5	0.2079	0.7285	0.8953	0.5328
7	0.1423	0.5819	0.9299	0.6897
9	0.1192	0.5386	0.9439	0.7286
11	0.0974	0.5146	0.9542	0.7359
13	0.0918	0.5220	0.9584	0.7337
14	0.0732	0.4365	0.9673	**0.7896**
15	0.0691	1.2210	0.9694	0.5347

PGD Detector Test Accuracy: 78.71% (Target: ≥70%)

3.2.2 BIM Detector Training

Table 18: BIM Adversarial Detector Training (ResNet-34)

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7093	0.6967	0.4974	0.5003
5	0.2475	2.8702	0.8880	0.5595
9	0.1151	0.3796	0.9521	0.8554
14	0.0508	0.3159	0.9799	0.8830
19	0.0497	0.2059	0.9816	**0.9165**
25	0.0236	16.7720	0.9915	0.5199

BIM Detector Test Accuracy: 91.48% (Target: ≥70%)

3.2.3 Detection Performance Comparison

Table 19: Adversarial Detection Results Summary

Attack Type	Test Detection Acc	Target	Status
PGD	78.71%	≥70%	
BIM	91.48%	≥70%	

Analysis:

- BIM detector achieves significantly higher accuracy (91.48%) compared to PGD (78.71%)
- This suggests BIM perturbations may be more distinguishable or have more consistent patterns
- Both detectors successfully meet the ≥70% requirement
- The high detection rates indicate that adversarial examples have detectable statistical differences from clean images

4 Conclusion

4.1 Q1 Summary

- Successfully fine-tuned ViT-S on CIFAR-100 with LoRA
- Best configuration (Rank=8, Alpha=8) achieved **89.33%** test accuracy
- LoRA improved accuracy by **+8.17%** over baseline (81.16% → 89.33%)
- Parameter efficiency: Only 1% of parameters trainable with LoRA
- Optuna found optimal hyperparameters: Rank=4, Alpha=4, LR=0.00048
- Partial freeze experiment showed full-freeze LoRA is more effective

4.2 Q2 Summary

- Clean ResNet18 achieved **83.26%** on CIFAR-10 (target: ≥72%)
- FGSM attacks reduced accuracy to 9.59% (scratch) and 19.06% (ART)
- PGD detector achieved **78.71%** detection accuracy
- BIM detector achieved **91.48%** detection accuracy